

CHANDAN SATAPATHY

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SUMMARY

Backend Engineer with 3+ years of experience at **Udaan.com** — India's largest B2B e-commerce platform, serving **3M+ retailers across 900+ cities**. As the **sole engineer** accountable for **Warehouse Management System** reliability and roadmap, designed and shipped distributed systems spanning inventory tracking, logistics dispatch, and asset reconciliation at scale. Reduced total infrastructure cost by ~ **20%**, improved picking throughput by **50%**, and cut P0 detection time by **40%** through database performance engineering, microservices architecture and observability. Proficient in **Kotlin, Python, TypeScript, Java** and cloud-native systems on **Azure & AWS**, with a strong foundation in data structures, algorithms, and system design.

WORK EXPERIENCE

UIX Labs

Mar 2026 - Present

Software Consultant

Remote

- Delivering backend architecture and feature development across multiple client projects spanning warehouse management and logistics.

Udaan.com

Aug 2022 - Sept 2025

Software Engineer 2

Aug 2024 - Sept 2025 | Bangalore

- Owned **Warehouse Management & Storage Systems (WMS)** powering Udaan's B2B supply chain across **20+ warehouses**, processing **millions of inventory events daily**.
- Mentored 2 graduate engineers over 12 months through weekly 1:1s and code reviews; both reached independent sub-system ownership within 6 months.
- Designed and owned **Double Dispatch - a parallelised last mile logistics system** - in Kotlin and TypeScript, enabling concurrent last-mile manifest generation across multiple route batches and improving dispatch throughput by ~**30%** without any additional infrastructure. Led cross-functional design sync with ops and logistics teams to align on dispatch SLAs before implementation.
- Diagnosed **parameter-sniffing and missing index root causes** behind query performance degradation and CPU spikes in Azure SQL and CosmosDB - profiled execution plans, redesigned indexes aligned to actual usage patterns, cutting **CPU requirements from 32 to 16 cores (50% cost reduction)**.
- Planned and executed the **Asset Receipt Note (ARN)** system with concurrency-safe distributed locks given write-heavy, multi-site access patterns; designed CosmosDB bulk upserts for idempotent tote reconciliation; schema supports eventual consistency across **20+ warehouse sites** with reconciliation SLAs.
- Contributed to the **Fresh vertical platform migration**, realigning **15+ legacy warehouse flows** with a new distributed architecture - achieving **50% improvement in ground picking times** and **80% increase in item accountability**.
- Designed idempotent data archival and entity demerger jobs migrating org-level inventory across microservices, reducing active table sizes by ~**40%** and cutting long-tail query latency on high-cardinality tables.
- Established SLO aligned error limits and built **Prometheus-based observability** across **4+ microservices** - 4xx/5xx alerts, Jetty request alerts, Slack pipelines - reducing mean time to detection (MTTD) for P0 incidents by ~**40%**.
- Implemented **retry logic, timeouts, and circuit-breaker patterns** across Picklist and archival workflows - reducing on-call escalations by ~**25% quarter-over-quarter**.

- Built an **anomaly detection framework** (Python, Prophet algorithm) integrated into the warehouse data pipeline, reducing system availability drops by **30% across 100K+ daily procurement records**.
- Resolved critical **CosmosDB throttling** by refactoring query patterns and batching bulk writes - reducing throttling events by 37% and cutting cloud DB costs on a high-frequency write workload.
- Designed **real-time operational dashboards** in React and TypeScript for ground teams, monitoring picking efficiency, failure rates, and warehouse health - **adopted across 5+ sites by 100+ ops users**.
- Expanded **unit and integration test coverage by 25%** across the WMS codebase - migrating **50+** test suites to JUnit 5, reducing production deployment regressions.

Enthire Co.**Jan 2021 - July 2021***Machine Learning Intern**Remote*

- Developed production ML models and REST APIs (Python, FastAPI), reducing prediction time lag by **80%** and improving accuracy from **~78% to 94%** via feature engineering with Scikit-Learn and TensorFlow.
- Built a **1-click AWS deployment module** from scratch - auto-provisioning FastAPI backends, containerizing with Docker, and rendering a Streamlit monitoring dashboard - reducing deployment setup from **4+ hours to under 5 minutes (98% reduction)**.

EDUCATION

National Institute of Technology, Kurukshetra**Aug 2018 - May 2022***B.Tech, Electrical Engineering**Kurukshetra, Haryana*

- **CGPA: 9.16/10**
- Competitive programmer; Electoreck Tech Club Core Member (4 years); Debate Club

Ryan International School**May 2018***Senior Secondary (PCM + Economics) - 94.4% | Secondary - 10/10 CGPA**Gurgaon, Haryana***SKILLS**

- **Language Stack:** Kotlin, Java, Python, JavaScript, TypeScript, Dart
- **Database Stack:** Azure SQL, CosmosDB, MongoDB, MySQL, Redis
- **Frameworks:** FastAPI, Flask, Django, Pyramid, Flutter, Terraform
- **Cloud Services Stack:** Microsoft Azure, AWS (Lambda, EC2, S3), Google Cloud Platform, Serverless
- **ML / Data Stack:** TensorFlow, Scikit-Learn, Pandas, NumPy, SciPy, Matplotlib, Docker
- **Core CS:** Distributed Systems, System Design, Data Structures & Algorithms, Concurrency, TDD, CI/CD

ACHIEVEMENTS

- **Google Hash Code 2021** - Global Rank **2,613** out of hundreds of thousands of participants worldwide.
- **HackerRank Algorithms** - Rank **3,210 / 3.4M+** (top 0.01% globally); CodeChef September Challenge 2020 - Global Rank **357 / 30,000**.

INTERESTS

Finance, economics & international affairs; competitive programming; avid reader of novels and mythology.